

Asynchronous Reward-Calibrated Peer Learning Architecture and Core Algorithm

Idea note | October 2, 2026

Core idea

Identical agents start from one initial language model, solve problems in separate environments, and perform independent GRPO updates. After each update, an agent broadcasts a frozen model snapshot. A receiver selects a peer from its current-round inbox and scores its own trajectories with that teacher. Outcome correctness and the teacher’s preference for the actual answer determine whether token-level peer signals are kept or reversed. The idea connects independent solving, asynchronous model exchange, and reward-calibrated top-4 on-policy distillation (OPD) into a continuing learning loop intended to improve mean population capability.

1 Architecture and policy roles

The system comprises a pool from one problem family, an external strong-model judge, and N independent agents. Each agent owns its environment, trajectories, trainable parameters, and receive cache. The pool supplies tasks and the judge supplies outcome labels. All effective policies share the same initialization:

$$\theta_a^{(0)} = \theta_0, \quad a \in \{1, \dots, N\}. \quad (1)$$

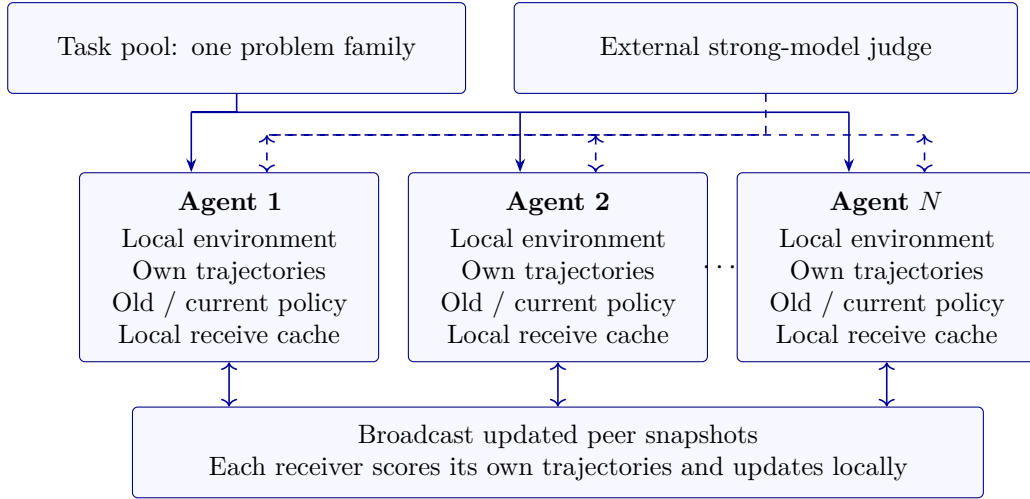


Figure 1: Logical architecture. The pool, judge, and broadcast channel are shared interfaces; solving, caching, and optimization remain local to each agent.

Three policy versions have distinct roles: π_{old} is the frozen behavior policy for the round, π_T is its locked peer teacher, and π_θ is the local policy being optimized. A broadcast snapshot supplies a learning signal and does not overwrite the receiver. Gradients update only local trainable parameters, which can be LoRA adapters on a frozen base.

2 Asynchronous coordination and cache protocol

2.1 Round-local receive window and teacher locking

Agent a advances local round b without global synchronization, clearing all leftover snapshots at its start. Entry e records its sender, sender-local update number, snapshot identity, and receiver-side arrival time $t_{a,e}^{\text{arr}}$. Let $t_{a,b}^{\text{start}}$ denote round start and $t_{a,b}^{\text{A}}$ completion of all rewards and GRPO advantages. The eligible set is:

$$\mathcal{C}_{a,b} = \{e : t_{a,b}^{\text{start}} \leq t_{a,e}^{\text{arr}} \leq t_{a,b}^{\text{A}}\}. \quad (2)$$

An empty set gives local GRPO plus reference KL, without peer OPD. Otherwise use the receiver's latest arrival, breaking ties uniformly among those latest entries:

$$\begin{aligned} t_{a,b}^{\text{max}} &= \max_{e \in \mathcal{C}_{a,b}} t_{a,e}^{\text{arr}}, \\ \mathcal{E}_{a,b} &= \{e \in \mathcal{C}_{a,b} : t_{a,e}^{\text{arr}} = t_{a,b}^{\text{max}}\}, \\ e_{a,b}^* &\sim \text{Unif}(\mathcal{E}_{a,b}), \quad \pi_{a,b}^T = \pi_{e_{a,b}^*}^{\text{snapshot}}. \end{aligned} \quad (3)$$

Arrivals at advantage completion are eligible. Lock the chosen teacher through optimization. Later arrivals are cached but cannot change the teacher, scores, or branch; delete them next round. Sender-local counts do not define global versions, and recency does not imply quality.

2.2 How agents progress concurrently

Each agent follows: roll out $\rightarrow$ judge and compute advantages $\rightarrow$ select and score a teacher $\rightarrow$ optimize $\rightarrow$ broadcast. All 128 trajectories finish before teacher selection:

$$t_{a,b}^{\text{roll}} \leq t_{a,b}^{\text{A}} = t_{a,b}^{\text{select}} \leq t_{a,b}^{\text{T,start}} \leq t_{a,b}^{\text{T}} \leq t_{a,b}^{\text{opt,start}}. \quad (4)$$

For agent a 's local batch b , the timestamps are:

Timestamp	Meaning
$t_{a,b}^{\text{roll}}$	Completion of all 128 rollouts in the batch
$t_{a,b}^{\text{A}}$	Completion of all outcome rewards and GRPO advantages
$t_{a,b}^{\text{select}}$	Logical instant when the peer teacher is selected and locked
$t_{a,b}^{\text{T,start}}$	Start of batch scoring by the locked teacher
$t_{a,b}^{\text{T}}$	Completion of all required teacher scores for the batch
$t_{a,b}^{\text{opt,start}}$	Start of local parameter optimization for this batch

Teacher selection occurs at advantage completion; with no teacher, skip scoring.

The local update waits for all teacher scores, without waiting for peers. Updating agents stop intake; other agents continue. If all agents update, intake pauses. Each completed update immediately broadcasts an immutable snapshot to every other cache, including joint and shutdown updates. Start a new round when intake permits. On shutdown, finish ongoing rollouts, scoring, and updates. Concurrency is across agents; the final teacher scores only after its batch rollouts finish.

3 Local trajectories, outcome rewards, and GRPO advantages

Each agent draws 16 tasks per round and generates 8 local trajectories for each task. Task index q , within-task rollout index k , and batch trajectory index i satisfy:

$$Q = 16, \quad G = 8, \quad B = QG = 128, \quad i = (q - 1)G + k. \quad (5)$$

A trajectory τ_i can include repeated code inspection, edits, tests, and repairs. Token y_{it} is a local policy action and h_{it} is its actual prefix, including the task, previous actions, and environment feedback. All rollouts in the batch use one π_{old} ; broadcasts do not change it. The teacher scores the same h_{it} rather than executing an alternative environment path.

Let z_i be the final answer and $\mathcal{V}(x_q, z_i)$ the external judge’s correctness label: 1 for correct and 0 for incorrect. The terminal reward is:

$$R_i = \begin{cases} +1, & \mathcal{V}(x_q, z_i) = 1, \\ -1, & \mathcal{V}(x_q, z_i) = 0. \end{cases} \quad (6)$$

GRPO compares the 8 trajectories for the same task, rather than mixing different tasks into one reward group. The mean reward and sample standard deviation are:

$$\bar{R}_q = \frac{1}{G} \sum_{k=1}^G R_{q,k}, \quad s_q = \sqrt{\frac{1}{G-1} \sum_{k=1}^G (R_{q,k} - \bar{R}_q)^2}. \quad (7)$$

The sample-standard-deviation denominator is $G - 1 = 7$. Only a zero standard deviation is replaced by a positive ϵ_{std} . The resulting trajectory advantage is shared across its local action tokens used by GRPO:

$$\tilde{\sigma}_q = \begin{cases} s_q, & s_q > 0, \\ \epsilon_{\text{std}}, & s_q = 0, \end{cases} \quad A_{q,k,t}^G = A_{q,k}^G = \frac{R_{q,k} - \bar{R}_q}{\tilde{\sigma}_q}, \quad \epsilon_{\text{std}} > 0. \quad (8)$$

If all 8 trajectories succeed or all fail, the numerator is also zero and the GRPO advantage is zero. The group stays in the batch; peer OPD and reference KL follow their own rules. F_i contains actual final-answer positions, and M_i^G , M_i^D , and M_i^K are the local action masks for the three losses. These are input interfaces to the equations. Environment observations enter the context but are not automatically optimized as action tokens.

4 Calibrating peer-signal direction with outcome correctness

After locking a peer teacher, the receiver evaluates its own actual final answer by teacher forcing. For nonempty F_i , compute the mean token log probabilities under the teacher and local old policy:

$$\bar{\ell}_i^T = \frac{1}{|F_i|} \sum_{t \in F_i} \log \pi_T(y_{it} | h_{it}), \quad \bar{\ell}_i^{\text{old}} = \frac{1}{|F_i|} \sum_{t \in F_i} \log \pi_{\text{old}}(y_{it} | h_{it}). \quad (9)$$

$$J_i = \bar{\ell}_i^T - \bar{\ell}_i^{\text{old}} = \frac{1}{|F_i|} \sum_{t \in F_i} \log \frac{\pi_T(y_{it} | h_{it})}{\pi_{\text{old}}(y_{it} | h_{it})}, \quad |F_i| > 0. \quad (10)$$

$J_i > 0$ means the teacher assigns greater support to this actual answer, and $J_i < 0$ means less support. Combine this comparison with the terminal reward to retain, reverse, or disable the trajectory’s peer signal:

$$\lambda_i = \begin{cases} 0, & |F_i| = 0, \\ +1, & |F_i| > 0 \wedge (R_i J_i > 0 \vee J_i = 0), \\ -1, & |F_i| > 0 \wedge R_i J_i < 0. \end{cases} \quad (11)$$

Outcome reward	J_i	λ_i	Peer-signal direction
+1	> 0	+1	Keep: teacher favors a correct answer
+1	< 0	-1	Reverse: teacher disfavors a correct answer
-1	< 0	+1	Keep: teacher disfavors an incorrect answer
-1	> 0	-1	Reverse: teacher favors an incorrect answer

For a nonempty answer with $J_i = 0$, set $\lambda_i = +1$ and retain ordinary peer guidance; a zero average does not imply zero differences at every token. For an empty final answer, set $\lambda_i = 0$ before computing J_i and explicitly return zero peer loss. The original GRPO group and advantages remain unchanged.

This calibration supplies a trajectory-level direction rather than stepwise correctness labels. J_i compares probabilities of the same local answer, not the teacher’s own solving accuracy. Reverse distillation reverses the receiver’s OPD advantage; it neither trains the teacher nor swaps the direction of KL.

5 Token-level peer advantages on fixed old-policy top-4 candidates

At every local trajectory prefix h_{it} , take only the behavior policy’s top-4 vocabulary tokens. Candidate identities remain fixed throughout scoring and optimization:

$$S_{it} = \text{Top4}(\pi_{\text{old}}(\cdot \mid h_{it})), \quad |S_{it}| = 4. \quad (12)$$

Top-4 means scoring four alternative tokens at an existing prefix; it does not generate four additional full trajectories. The peer teacher scores those candidates, and the current local policy recomputes their probabilities during optimization. For $v \in S_{it}$:

$$\ell_{itv}^{\text{old}} = \log \pi_{\text{old}}(v \mid h_{it}), \quad \ell_{itv}^T = \log \pi_T(v \mid h_{it}), \quad \ell_{itv}^\theta = \log \pi_\theta(v \mid h_{it}). \quad (13)$$

All three probabilities come from a full-vocabulary softmax. Only the candidate weights w_{itv} are renormalized within top-4. Teacher–student differences and current–old policy ratios still use full-vocabulary probabilities:

$$w_{itv} = \frac{\exp(\ell_{itv}^{\text{old}})}{\sum_{u \in S_{it}} \exp(\ell_{itu}^{\text{old}})}, \quad \sum_{v \in S_{it}} w_{itv} = 1. \quad (14)$$

Symmetrically clip the teacher–old log-probability difference, then apply the trajectory-level calibration sign and candidate weight:

$$d_{itv} = \ell_{itv}^T - \ell_{itv}^{\text{old}}, \quad \Delta_{itv} = \text{clip}(d_{itv}, -C, C), \quad C > 0. \quad (15)$$

$$\boxed{a_{itv}^{\text{peer}} = \text{sg}[\lambda_i w_{itv} \Delta_{itv}]}, \quad \lambda_i \in \{-1, +1\}. \quad (16)$$

sg denotes stop-gradient. The old policy, teacher, candidate identities, weights, differences, and λ_i are fixed training signals. A sign of +1 retains guidance, −1 reverses it, and 0 uses an explicit zero-loss branch. Gradients flow through current-policy probabilities into local trainable parameters only.

Scoring data must include log probabilities for actual answer tokens as well as top-4 candidates. A sampled token may fall outside top-4 but is still needed for J_i . Candidate losses are summed without another division by four because their weights already sum to one.

6 Separately clipped objectives and local optimization

OPD uses candidate-token ratios, while GRPO uses sampled-token ratios. Both compare the current local policy with the old local policy; neither places the teacher in the denominator:

$$\rho_{itv}(\theta) = \frac{\pi_{\theta}(v \mid h_{it})}{\pi_{\text{old}}(v \mid h_{it})}, \quad r_{it}(\theta) = \frac{\pi_{\theta}(y_{it} \mid h_{it})}{\pi_{\text{old}}(y_{it} \mid h_{it})}. \quad (17)$$

$$c_X(r) = \text{clip}\left(r, 1 - \epsilon_X^{\text{low}}, 1 + \epsilon_X^{\text{high}}\right), \quad X \in \{\text{G}, \text{D}\}. \quad (18)$$

The two PPO clipped-surrogate losses are:

$$L_{it}^{\text{G}} = -\min\{r_{it}A_i^{\text{G}}, c_{\text{G}}(r_{it})A_i^{\text{G}}\}. \quad (19)$$

$$\begin{aligned} U_{it}^{\text{peer}} &= -\sum_{v \in S_{it}} \min\{\rho_{itv}a_{itv}^{\text{peer}}, c_{\text{D}}(\rho_{itv})a_{itv}^{\text{peer}}\}, \\ L_{it}^{\text{peer}} &= \begin{cases} 0, & \lambda_i = 0, \\ U_{it}^{\text{peer}}, & \lambda_i \in \{-1, +1\}. \end{cases} \end{aligned} \quad (20)$$

For each branch, first average token losses within each trajectory using its action mask, then average equally over the $B = 128$ trajectories in the batch and combine:

$$\mathcal{R}_M(X) = \frac{1}{B} \sum_{i=1}^B \left(\frac{1}{|M_i|} \sum_{t \in M_i} X_{it} \right), \quad |M_i| > 0 \quad (i = 1, \dots, B). \quad (21)$$

$$\boxed{\mathcal{L}_{\text{total}} = \alpha \mathcal{R}_{\text{MG}}(L^{\text{G}}) + \beta \mathcal{R}_{\text{MD}}(L^{\text{peer}}) + 0.01 \mathcal{L}_{\text{KL}}}. \quad (22)$$

α and β scale GRPO and peer OPD. Every trajectory receives equal batch weight regardless of length. Zero OPD trajectory contributions from empty answers remain in the denominator B ; do not renormalize over trajectories with nonzero λ_i . GRPO, peer OPD, and reference KL all use this reduction. Without an eligible teacher, use local GRPO and reference KL. The independent reference-policy regularizer $\mathcal{L}_{\text{KL}}$ has coefficient 0.01 and is not multiplied by λ_i .

Reverse the advantage before min/clip; do not negate an already-clipped ordinary OPD loss. Indeed:

$$\begin{aligned} \ell(r, -a) &= \max\{ra, c(r)a\}, \\ -\ell(r, a) &= \min\{ra, c(r)a\}, \quad \ell(r, a) = -\min\{ra, c(r)a\}. \end{aligned} \quad (23)$$

Likewise, clip the branches separately before combining them; do not add their advantages and clip once. PPO clipping is used without introducing a separate PPO critic or an additional historical-trajectory update.

7 End-to-end loop and method rationale

7.1 Independent event loop for each agent

1. Start a round: clear the old cache, freeze π_{old} , draw 16 tasks, and collect 8 trajectories per task. Append incoming peer broadcasts to this round’s cache.
2. Obtain outcome rewards; compute the reward mean, sample standard deviation, and A_i^G within each group of 8 trajectories.
3. At advantage completion, include simultaneous arrivals. If no peer is eligible, proceed locally; otherwise select the latest arrival, break ties uniformly, and lock the teacher.
4. If a teacher was selected, score the entire batch of local prefixes. Compute J_i, λ_i on actual answers and $w_{itv}, \Delta_{itv}, a_{itv}^{\text{peer}}$ on fixed old-policy top-4 candidates.
5. Optimize the current local policy: clip GRPO and OPD separately, average tokens within each trajectory and then average equally across batch trajectories before combining and adding reference KL. Do not update the old policy or teacher.
6. Immediately broadcast the completed update, then start another round when intake permits. During shutdown, finish ongoing work and broadcast completed updates as usual.

7.2 Reference-KL interface

The reference policy π_{ref} has a distinct role from the peer teacher and supplies reference scores only. The shared k3 / low-var core and its batch reduction can be written as:

$$\begin{aligned}\delta_{it}(\theta) &= \log \pi_{\text{ref}}(y_{it} \mid h_{it}) - \log \pi_{\theta}(y_{it} \mid h_{it}), \\ K_{it}^{(3)}(\theta) &= \exp(\delta_{it}(\theta)) - 1 - \delta_{it}(\theta), \\ \mathcal{L}_{\text{KL}} &= \mathcal{R}_{M^{\text{K}}} \left(g_{\text{KL}} \left(K_{it}^{(3)}(\theta) \right) \right).\end{aligned}\tag{24}$$

g_{KL} encapsulates the reference-KL numerical implementation. The reference policy, action masks, and training hyperparameters are algorithm inputs rather than configuration details of this idea note. This term differentiates through the current local policy, is not detached as a whole, and is not automatically disabled by an empty final answer.

7.3 Central research proposition

The formulation draws on GRPO, PPO clipping, and OpenClaw-RL’s top-k OPD structure. It asks whether asynchronous independent learners can use peer snapshots to construct outcome-calibrated auxiliary advantages on their own trajectories. Model knowledge is exchanged, while training trajectories remain local and on-policy for each receiver. The trajectory-level λ_i does not certify every action or guarantee monotonic accuracy improvements. This note specifies a mechanism and hypothesis, not an empirical result.